1. On the Formation of the form System of the Ternary Biquadratic form

Reports of the Physical-Medical Society in Erlangen 39 (1907), pp. 176–179

(Extract from the author's dissertation.)

Work by Gordan, Maisano, and Pascal[1] deals with the form system of the ternary biquadratic form. Gordan sets up the complete form system, consisting of 54 formations, for the special automorphic form

$$f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1,$$

on the basis of principles similar to those which he had given for form systems in the binary domain.

In Maisano's work, for the general biquadratic form, the forms up to and including the fifth order[2] are set up, together with some invariants, covariants, and contravariants of higher order, according to the method used by Gordan in volume I of the *Mathematische Annalen* for the ternary cubic form. Pascal, using Maisano's results, is chiefly concerned with the question of the decomposition of the biquadratic form into factors[3].

The aim of my investigations is to set up the form system for the general ternary biquadratic form; to begin with, only the main foundations are given, and a so-called "relatively complete system"[4] is established.

The basic idea for forming systems of ternary forms is the same as in the binary domain. Starting from a first relatively complete system – the system of forms taken over from the binary case – one passes, according to a definite law, to systems with ever higher modulus, until either the system of a modulus – taking the modulus as ground form – becomes finite and known, or until a modulus can be reduced to forms having invariants as factors. By transvection of the relatively complete system with the system of the modulus, in the first case one obtains the absolutely complete system, while in the second case the relatively complete and absolutely complete systems coincide. By Hilbert's general proof the finiteness of form systems, this procedure must necessarily come to an end.

In our case, the modulus (abc) of the first relatively complete system can be led back to the moduli

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2 \quad \text{and} \quad \nu = (abu)^4;$$

from this the relatively complete system modulo ν is readily obtained. As the sequence of moduli we now choose the forms

$$\nu; \qquad \nu^{(\nu)} = (\nu\nu_1 x)^4 = s_x^4,$$
$$\nu^{(s)} = (ss'u)^4 \ldots,$$

and adjoin, as further moduli, two quadratic forms occurring in the formation of the relatively complete system modulo s, namely u_σ^2 and t_x^2. One can then show that the modulus following s, namely $(ss'u)^4$, is reducible to the modulus (ϱ, t). Since, however, the simultaneous system of two quadratic forms is finite

[1] P. Gordan, "Über the volle Formensystem of the ternären biquadratischen form" $f = x_1^3x_2 + x_2^3x_3 + x_3^3x_1$ (Math. Annalen vol. XVII, pp. 217–233, 1880); G. Maisano, 1. "Sistemi completi dei primi cinque gradi della forma ternaria biquadratica e degl' invarianti, covarianti e contravarianti di sesto grado" (Giorn. di Battaglini XIX). 2. "Sui covarianti indipendenti di 6° grado nei coefficienti della forma biquadratica ternaria" (Rend. Circ. Mat. di Palermo I, 1887); E. Pascal, "Contributo alla teoria della forma ternaria biquadratica e delle sue varie decomposizioni in fattori" (Memoria premiata dalla R. Accademia delle scienze fisiche e matematiche di Napoli, 1905).

[2] By "order" the dimension in the coefficients is to be understood, and by "degree" the dimension in the variables.

[3] In his second short note Maisano attempts to prove a linear dependence among the three covariants of order 6 and degree 6. In contrast to this not quite complete proof, Pascal believes he has proved the linear independence of the three covariants of order 6, as well as that of the three invariants of order 9. That, however, a linear relation does in fact exist between the three covariants on the one hand and the three invariants on the other, emerged in the course of my computations; in Pascal's notation the explicit formulas are

$$0 = 20\Omega_1 + 6\Omega_2 - 3\Omega_3 + 4fC_1 - 12fC_2 - 4A \cdot \Delta,$$

$$0 = 4A^3 - 15AB + 30C - 90D + 9E.$$

[4] For the terminology, see Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie*, vol. II, p. 227.

and known, the termination described above has thereby been reached. As the relatively complete system modulo (ϱ, t) one obtains 331 formations. —

I add a few details concerning the method used.

The fundamental process for producing forms, the folding process, can be defined in the ternary domain as follows:

If in a symbolic product

$$s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$$

one replaces the pairs of factors

	$s_x t_x$	$u_\sigma u_\tau$	$s_x u_\sigma$ or $s_x u_\tau$	$t_x u_\tau$ or $t_x u_\sigma$
respectively by	(stu)	$(\sigma\tau x)$	s_σ or s_τ	t_τ or t_σ
folding	I	II	III	IV,

then the forms so arising have been obtained from the original form by folding[5].

Here one may still always put $s_\sigma = 0;\ t_\tau = 0$. For the relation among the individual foldings, the following theorem holds:

The foldings I and II are fundamental foldings, from which, independently of the order of composition, the foldings III and IV can be composed. In other words: to form all forms arising from a given expression by folding, one need only apply foldings I and II[6].

By a "form sequence" we mean an initial form together with all forms arising from it by folding into itself. By the theorem, a form sequence $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$ can be arranged in a rectangular scheme. In proceeding, one obtains:

1. by moving one column to the right, all forms arising by folding I from the adjacent forms;

2. by moving one row downward, all forms arising by folding II from the forms standing above, and hence all forms arising by folding.

Under these assumptions, the theorems on reducers can be stated in their most general form, under the definition

"A reducer is a reducible form sequence,"

as follows:

If the initial form of a form sequence is reducible because one of its terms has arisen by folding with a reducer, and if the final form of the form sequence has arisen from this same term by folding, then the whole form sequence is reducible.

Further reduction methods to be mentioned are:

1. the so-called "double reduction," that is, the reduction of a form in two ways in order to obtain a relation between the higher forms;

2. "folding with decomposed forms," which partly has the character of reduction by reducers and partly that of "double reduction."

[5] Gordan, Math. Annalen, vol. XVII, p. 219.

[6] The theorem has an exception in the case where n and μ, respectively m and ν, vanish simultaneously; the "form sequence" then reduces to the diagonal term.